GATE-2005
BT: BIOTECHNOLOGY

Q.1 – 10 carry one mark each

1. Cells of meristemoid are best described as
   (a) differentiated and non dividing
   (b) dedifferentiated and dividing
   (c) differentiated and dividing
   (d) dedifferentiated and non dividing
2. Ultrafiltration process can not be used for
   (a) fractionation of proteins
   (b) desalting
   (c) harvesting of cells
   (d) selective removal of solvents
3. The number of replicons in a typical mammalian cell is
   (a) 40-200 (b) 400
   (c) 1000-2000 (d) 50000-100000
4. What product will result from complete hydrolysis of soluble dextran ?
   (a) Sucrose only
   (b) Fructose only
   (c) Glucose and fructose only
   (d) Glucose only
5. Aeration in a biorector is provided by
   (a) impeller (b) baffles
   (c) sparger (d) all of these
6. The transplastomic plants bear no risk for gene transfer through pollens as
   (a) the pollens degenerate before fertilization
   (b) the transformed mitochondrial DNA is lost during pollen maturation
   (c) the transformed chloroplast DNA is lost during pollen maturation
   (d) the transformed genomic DNA are inherited maternally
7. The mobility of DNA in agarose gel electrophoresis is solely based on its
   (a) charge
   (b) conformation
   (c) size
   (d) none of these
8. Which of the following fluorescent probes is used to monitor the progress of ampli fication in Real time PCR ?
   (a) SYBR green (b) Rhodamine
   (c) FITC (d) Cyan blue
9. Expression of which of the following reporter genes does not require addition of specific substrate for detection ?
   (a) Luciferase
   (b) β-Glucuronidase
   (c) β-Glucosidase
   (d) Green fluorescent protein
10. Cibacren Blue dye affinity chromatography can be used for affinity purification of
    (a) NADPH dehydrogenase
    (b) glucoamylase
    (c) subtilisin
    (d) caspase

Q. 11 – 26 carry two marks each

11. A linear DNA fragment is 100% labeled at one end and has 3 restriction sites for EcoRI. If it is partially digested by EcoRI so that all possible fragments are produced, how many of these fragments will be labeled and how many will not be labeled ?
    (a) 4 labeled; 6 unlabeled
    (b) 4 labeled; 4 unlabeled
    (c) 3 labeled; 5 unlabeled
    (d) 3 labeled; 3 unlabeled
12. Match the following products with their starting substrates

A. Sake	1. apple juice
B. cider	2. grape juice
C. wine	3. barley
D. lager	4. rice

    (a) A → 4, B → 1, C → 2, D → 3
    (b) A → 1, B → 4, C → 2, D → 3
    (c) A → 2, B → 3, C → 1, D → 4
    (d) A → 3, B → 4, C → 2, D → 1

13. Identify the following antibiotics with their modes of action.

Antibiotic	Mode of action
A. Ampicillin	1. inhibition of protein synthesis
B. Tetracycline	2. inhibition of cell wall synthesis
C. Nystatin	3. damage to cytoplasmic membrane
D. Anthramycin	4. damage to DNA structure

(a) A → 1, B → 2, C → 4, D → 3
(b) A → 2, B → 1, C → 3, D → 4
(c) A → 1, B → 2, C → 3, D → 4
(d) A → 3, B → 4, C → 2, D → 1

14. In a bioreactor baffles are incorporated to
(a) prevent vortex and to improve aeration efficiency
(b) maintain uniform suspension of cells
(c) minimize the size of air bubble for greater aeration
(d) maintain uniform nutrient medium

15. Somatic embryo from cotyledon explant would develop in the following sequential stages,
(a) cotyledonary → heart → globular → torpedo
(b) globular → torpedo → heart → cotyledonary
(c) globular → heart → torpedo → cotyledonary
(d) cotyledonary → globular → heart torpedo

16. Though the right border (RB) and left border (LB) of T-DNA are identical, the DNA transfer is specific for the DNA left of the RB (the T-DNA), rather than for the DNA left of the LB because
(a) the sequence context at the RB defines the direction of transfer
(b) the sequence context at the LB defines the direction of transfer
(c) the nuclear location sequence (NLS) of VirD2 protein drives the excised T-strand
(d) the endonuclease activity of VirD2 protein allows nicking at RB

17. Determine the correctness or otherwise of the following Assertion[a] and Reason [r] *Assertion:* An antigen recognized by one immunoglobulin subtype is not recognized by any other subtype.
Reason: Immunoglobulin subtypes differ from each other body in the variable and in the constant regions.
(a) Both [a] and [r] are true and [r] is the correct reason for [a]
(b) Both [a] and [r] are true but [r] is not the correct reason for [a]
(c) Both [a] and [r] are false
(d) [a] is true but [r] is false

18. Identical sized RNA transcript is detected by Northern blot analysis of UDP glucuronosyl transferase obtained from human liver and kidney. Microarray analysis of the same samples shows equal spot intensity, ohereas Western blot detects a 55 kDa strong band in liver, but a very faint band in kidney of same size. The regulation of UDP glucuronosyl transferase is
(a) transcriptionally controlled
(b) post-transcriptionally controlled
(c) translationally controlled
(d) post-translationally controlled

19. Match the items on the left column with those on the right

Left	Right
P. Programmed cell death at site of infection	1. TMV coat protein
Q. Hormone upregulated during flooding stress	2. EPSP synthase
R. Target for herbicide glyphosate	3. Hyper-sensitive response
S. Pathogen-derived resistance	4. Ethylene

(a) P-1, Q-2, R-4, S-3 (b) P-3, Q-4, R-2, S-1
(c) P-1, Q-4, R-2, S-3 (d) P-3, Q-2, R-4, S-1

20. Using the Hill equation for an enzyme $[S]_0 = (v_0 K_m / V_{max} - v_0)^{1/n}$ and the plot of $\log_{10} (v_0/V_{max} - v_0)$ *vs* log 10 $[S]_0$ one can find out
(P) V_{max} form the intercept on the ordinate
(Q) K_m form the intercept on the ordinalte
(R) '*n*' from the slope
(S) K_m from the intercept on the abscissa
(a) P, Q
(b) Q, R
(c) R, S
(d) P, S

21. Expression in poor amount and in inactive form of cDNA of a euharyotic protein in Escherichia coli using its expression vector is due to
(P) the absence of capping mechanism of mRNA
(Q) codon bias
(R) absence of polyadenylation
(S) absence of proper glycosylation
(a) P, Q (b) Q, R
(c) Q, S (d) P, S

Common Data for Questions 22, 23, 24:

A recombinant SV40 virus delivers c-myc cDNA, which has a unique *sal* I site, into muscle cells. Southern analysis of *Sal* I digested total genomic DNA of the muscle cell using c-myc cDNA porbe generates a smear.

22. The DNA smear obtained on Southern blot is due to

(*a*) head to head concatamer of viral DNA

(*b*) head to tail concatamer of viral DNA

(*c*) tail to tail concatamer of viral DNA

(*d*) random integration of viral DNA

23. Western blot analysis of c-myc expression of such transformed cells last for

(*a*) transiently

(*b*) upto five generations

(*c*) upto 10 generations

(*d*) more than 100 generations

24. Which of the following types of cancer will be observed in such transformed cells ?

(*a*) Adenoma (*b*) Melanoma

(*c*) Sarcoma (*d*) Hepatoma

Common Data for Questions 25, 26:

Normal primary hepatocytes can be artificially immortalized. Certain spontaneous mutants of immortalized hepatocytes are sensitive to ionizing radiation.

25. Which of the following genes are involved in Immortalization of primary hepatocytes?

(*a*) Telomerase and Cyclin D

(*b*) NFκB and Thymidine kinase

(*c*) Cyclin D and myc

(*d*) Telomerase and Ras

26. What would happen to the mutant cells by ionizing radiation?

(*a*) Apoptosis

(*b*) Necrosis

(*c*) Cell growth arrest

(*d*) Cell proliferation

Linked Answer Questions: Q 27 a to Q 28b carry two marks each

Statement for Linked Answer Questions 27a & 27b

An aliquot of competent *E. coli* cells were used for determination of cell density by plate count method and another aliquot was used for transformation by plasmid DNA

27a. *E. coli* cell culture (1 ml) was diluted 1:1000000 and 200μl of this was used for plating. After 12h incubation of the plate, the number of colony forming units (CFU) was 150. What is the total CFUper ml in original culture ?

(*a*) 7.5×10^8 (*b*) 1.5×10^8

(*c*) 1.5×10^6 (*d*) 3.0×10^6

27b. Isolated plasmid DNA (5μg) was used for transformation of 100 μl competent *E. coli* cells to which 900 μl of SOC medium was added. An aliquot of 50ul was plated on a selective plate. After overnight incubation, 300 colonies were observed. Calculate the efficiency of transformation and the percentage of transformed cells per ml of parent culture

(*a*) 6.0×10^5 colonies per μg of plasmid DNA, 0.01%

(*b*) 1.2×10^5 colonies per μg of plasmid DNA, 0.02%

(*c*) 1.2×10^6 colonies per μg of plasmid DNA, 0.008%

(*d*) 6.0×10^6 colonies per μg of plasmid DNA, 0.01%

Statement for Linked Answer Questions 28a & 28b.

HMGCoA reductase that binds HMGCoA, is the major rate limiting step in the cholesterol biosynthetic pathway. Several inhibitors of this enzyme are used as potential drugs. The assay of the enzyme is based on labeling the enzyme with radiolabeled HMGCoA and counting (cpm) the labeled enzyme substrate complex in the presence (test) and in the absence (control) of the inhibitor. A bland is set up that contains no enzyme.

28a. The percent inhibition for this enzyme is calculated from the equation

(*a*) {[cpm (control) – cpm (test)]/[cpm (control) – cpm (blank)]} × 100

(*b*) {[cpm (control) – cpm (test)]/[cpm(blank) – cpm (control)]}} × 100

(*c*) {[cpm (test) – cpm (control)]/[cpm (control) – cpm (blank)]}} × 100

(*d*) {[cpm (control) – cpm (blank)]/[cpm (test) – cpm (control)]}} × 100

28b. An inhibitor is considered active if it causes more than 65% inhibition. The. cpm Values respectively of control, test and blank samples for inhibitors W, X, Y and Z are given below. State which of the inhibitors is active.

(*a*) X – 8000, 4000 and 100

(*b*) W – 7000, 1400 and 135

(*c*) Y – 7500, 5000 and 90

(*d*) Z – 7200, 2800 and 200

ANSWERS

1. (c)	**2.** (c)	**3.** (d)	**4.** (d)	**5.** (c)	**6.** (d)	**7.** (c)	**8.** (a)	**9.** (a)	**10.** (a)
11. (c)	**12.** (a)	**13.** (b)	**14.** (a)	**15.** (c)	**16.** (a)	**17.** (c)	**18.** (b)	**19.** (b)	**20.** (d)
21. (d)	**22.** (b)	**23.** (c)	**24.** (c)	**25.** (d)	**26.** (c)	**27.a** (a)	**27b.** (b)	**28a.** (a)	**28b.** (b)

EXPLANATIONS

1. The meristematic cells give rise to various organs of the plant, and keep the plant growing. The Shoot Apical Meristem (SAM) gives rise to organs like the leaves and flowers. The cells of the apical meristems - SAM and RAM (Root Apical Meristem) - divide rapidly and are considered to be indeterminate, in that they do not possess any defined end fate. In that sense, the meristematic cells are frequently compared to the stem cells in animals, which have an analogous behavior and function.

2. Ultrafiltration (UF) is a variety of membrane filtration in which hydrostatic pressure forces a liquid against a semipermeable membrane. Suspended solids and solutes of high molecular weight are retained, while water and low molecular weight solutes pass through the membrane. This separation process is used in industry and research for purifying and concentrating macromolecular (10^3 - 10^6 Da) solutions, especially protein solutions. Ultrafiltration is not fundamentally different from microfiltration, nanofiltration or gas separation, except in terms of the size of the molecules it retains. Ultrafiltration is applied in cross-flow or dead-end mode and separation in ultrafiltration undergoes concentration polarization.

3. A replicon is a DNA molecule, or a region of DNA or RNA, that replicates from a single origin of replication. There are 50000–100000 replicons in a typical mammalian cell.

4. Complete hydrolysis of dextran yield solely β-dglucose i.e., hydrolysis is accompanied by inversion at carbon-1 in such a way that new reducing ends are released only in the β-configuration.

5. Air is provided through sparger in bioreactors.

6. A transplastomic plant is a genetically modified plant in which the new genes have not been inserted in the nuclear DNA but in the DNA of the chloroplasts. The major advantage of this technology is that in many plant species plastid DNA is not transmitted through pollen, which prevents gene flow from the genetically modified plant to other plants.

7. The separation of the fragments in agarose gel is accomplished by exploiting the mobilities with which different sized molecules are able to pass through the gel. Longer molecules migrate more slowly because they experience more resistance within the gel. Because the size of the molecule affects its mobility, smaller fragments end up nearer to the anode than longer ones in a given period.

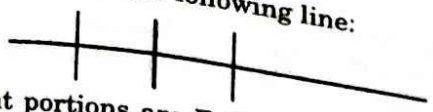
8. Real-time quantitative reverse transcriptase–polymerase chain reaction (RT–PCR) is the method of choice for rapid and reproducible measurements of cytokine or growth factor expression in small samples. Fluorescence detection methods for monitoring real-time PCR include fluorogenic probes labelled with reporter and quencher dyes, such as Taqman probes or Molecular Beacons and the dsDNA-binding dye SYBR Green I.

SYBR Green binds to the minor groove of the dsDNA only. In the solution the unbound dye exhibits little fluorescence. This fluorescence is enhanced when the dsDNA is bound with dye.

9. Luciferase bioluminescence color can vary between yellow-green (λ_{max} = 550 nm) to red (λ_{max} = 620) There are currently several different mechanisms describing how the structure of luciferase affects the emission spectrum of the photon and effectively the color of light emitted. One mechanism proposes that the color of the emitted light depends on whether the product is in the keto or enol form. The mechanism suggests that red light is emitted from the keto form of oxyluciferin, while green light is emitted from the enol form of oxyluciferin.

10. Cibacren blue dye (ligand) used in affinity chromatography, due to presence of sulfonated polyaromatic compounds which bind with considerable specificity and significant affinity to nucleotide-dependent enzymes and to a series of other proteins is used for glucoamylse purification.

11. Let the DNA be the following line:



The cut portions are E.CoRI digestion site. So the partial digestion can generate 3 smaller fragments with label and 5 fragments without any level.

12. Sake or saké is an alcoholic beverage of that is made from fermented rice. Cider is a fermented alcoholic beverage made from fruit juice, most commonly and traditionally apple juice. Grape juice is obtained from crushing and blending grapes into a liquid. The juice is fermented and made into wine, brandy or vinegar. Barley is used as a source of fermentable material for beer called lager and certain distilled beverages.

13. Ampicillin inhibits the third and final stage of bacterial cell wall synthesis in binary fission, which ultimately leads to cell lysis.

Nystatin forms pores in the membrane that lead to K+ leakage and death of the fungus. Tetracycline antibiotics are protein synthesis inhibitors, inhibiting the binding of aminoacyl-tRNA to the mRNA-ribosome complex.

Anthramycin exerts its biological activity through covalent binding to the C2-NH2 of guanine within the minor groove of DNA thus damaging the actual structure.

14. The main roles of baffles are prevent vertex formation, improve aeration and help in mixing.

15. In the globular stage, the embryo develops radial patterning through a series of cell divisions, with the outer layer of cells differentiating into the 'protoderm.' The globular embryo can be thought of as two layers of inner cells with distinct developmental fates; the apical layer will go on to produce cotyledons and shoot meristem, while the lower layer produces the hypocotyl and root meristem. Bilateral symmetry is apparent from the heart stage; provascular cells will also differentiate at this stage. In the subsequent torpedo and cotyledonary stages of embryogenesis, the embryo completes its growth by elongating and enlarging.

16. virulence genes are grouped genes for synthesis of opines and plant hormones are contained on the T-DNA; T-DNA is bordered by 25 bp direct repeats; only the TR border is required for transfer although the Ti border increased the efficiency when it is present; no sequences other that the borders are required for transfer.

17. The immunoglobulins can be divided into five different classes, based on differences in the amino acid sequences in the constant region of the heavy chains. The valency of Immunoglobulin refers to the number of antigenic determinants that an individual immunoglobulin molecule can bind. The valency of all immunoglobulins is at least two and in some instances more than that.

18. Regulation of UDP glucuronosyl transferase is the result of differential UGT gene expression and this is in response to those chemicals that are primarily eliminated by conjugation with glucuronic acid and hence more evident in liver and hence it can be stated that its regulation is post transcriptionally controlled.

19. Hypersensitive response is characterized by the rapid death of cells in the local region surrounding an infection carried out as a programmed mechanism for cell death. Environmental cues such as flooding, drought, chilling, wounding, and pathogen attack induce ethylene formation in plants. Glyphosate is a chemical herbicide which kills plants by inhbiting the shikimate pathway. It targets EPSP synthase, the enzyme that catalyzes the conversion of shikimate-3-phosphate and phosphoenolpyruvate into EPSP.

20. V_m can be plotted and calculated from the intercept on the ordinate while K_m will be given by Hill equation from the intercept of the abscissa.

21. The process of 52 capping is vital to creating mature messenger RNA, which is then able to undergo translation. Capping ensures the messenger RNA's stability while it undergoes translation in the process of protein synthesis. Glycosylation confers stability to the proteins and is also required for the folding of proteins. Absence of either of these mechanisms result in poor expression and inactivity.

22. The replication intermediates of parvoviruses thus are concatemers of head-to-head or tail-to-tail structure. Surprisingly, in case of the novel human bocavirus, neither head-to-head nor tail-to-tail DNA sequences were detected in clinical isolates; in contrast head-to-tail DNA sequences were identified by PCR and sequencing. Thereby, the head-to-tail sequences were linked by a novel sequence which can be detected by southern blot.

23. The Myc family of genes are proto-oncogenes implicated in cancer. Yamanaka et al. and Jaenisch et al. demonstrated that c-myc is a factor implicated in the generation of mouse iPS cells

and Yamanaka et al. demonstrated it was a factor implicated in the generation of human iPS cells. However, Thomson et al., Yamanaka et al., and unpublished work by Johns Hopkins University reported that c-myc was unnecessary for generation of human iPS cells. Usage of the "myc" family of genes in induction of iPS cells is troubling for the eventuality of iPS cells as clinical therapies, as 25% of mice transplanted with c-myc-induced iPS cells developed lethal teratomas. N-myc and L-myc have been identified to induce instead of c-myc with similar efficiency. C-Myc cells after transformation last for 10 generations.

24. Sarcoma is a cancer that arises from transformed cells of mesenchymal origin. Thus, malignant tumors made of cancerous bone, cartilage, fat, muscle, vascular, or hematopoietic tissues are, by definition, considered sarcomas. This is in contrast to a malignant tumor originating from epithelial cells, which are termed carcinoma. Sarcomas are quite rare. Common malignancies, such as breast, colon, and lung cancer, are almost always carcinoma.

25. Telomerase contains a catalytic unit with reverse transcriptase activity and an RNA part that provides template for telomere extension. Telomerase is normally expressed only in stem cells such as those found in hematopoietic tissues and germline cells but is nearly absent in most somatic cells. Induction of telomerase synthesis bypasses normal cellular **senesce** in cancer cells and endows them with unlimited replicative potential.

26. Ionizing radiation is generally harmful and potentially lethal to living things but can have health benefits in radiation therapy for the treatment of cancer and thyrotoxicosis. Its most common impact is the induction of cancer with a latent period of years or decades after exposure. High doses can cause visually dramatic radiation burns, and/or rapid fatality through acute radiation syndrome. Controlled doses are used for medical imaging and radiotherapy. So scientists suspect that low doses may have a hormetic effect that can improve health.

27.a Since, 1 ml = 1000 µl

New volume

$$= 10^3 \times 10^6\ \mu l = 10^9\ \mu l.$$

Since, only 200 µl is used we have a dilution fac of

$$= \frac{10^9\ \mu l}{200\ \mu l} = 0.5 \times 10^7$$

Number of colony in original one ml culture wo be

$$= 0.5 \times 10^7 \times 150$$

$$= 7.5 \times 10^8.$$

28.a. The amount of radioactivity will be in order control >test > blank

$$\%\ \text{inhibition} = \frac{\text{control} - \text{tes}}{\text{control} - \text{blan}}$$

decrease in enzyme complex

$$= \frac{\text{formation due to inhibitor}}{\text{maximum enzyme- substrate complex formation in absence of inhibitor}}$$

28.b. For each a, b, c, and d options respectively. The value of % inhibition is

For, $a = \dfrac{8000 - 4000}{8000 - 100} = 50.6\%$

$b = \dfrac{7000 - 1400}{7000 - 135} = 81.5\%$

$c = \dfrac{7500 - 5000}{7500 - 90} = 33.7\%$

$d = \dfrac{7200 - 2800}{7200 - 200} = 62.86\%$